

Derbyshire Shared Care Pathology Guidelines

Viral Hepatitis Serology

Purpose of guideline

This guideline gives guidance on the appropriate requesting of viral hepatitis serology in primary care and the interpretation of results. It is important to consider a viral cause of hepatitis when a patient presents with a new unexplained acute hepatitis or raised ALT, as it is key to prevent onward transmission. Knowing which hepatitis virus is responsible will inform onward management of the patient and their contacts. Post exposure prophylaxis may also be necessary and Public Health England notification and involvement. Most viral causes are self-limiting but few can cause severe disease, chronic infection and transmission during pregnancy.

Hepatitis A virus (HAV)

- Highly contagious, transmitted by faeco-oral contact
- Usually contracted by ingesting food or water contaminated with the virus or by coming in contact with an infected person (men who have sex with men (MSM))
- Infection is usually mild but the virus can rarely cause severe, acute disease
- Hepatitis A does not cause a chronic infection, like hepatitis B and C can

Hepatitis B virus (HBV)

- The most common cause of acute viral hepatitis worldwide
- Spread through contact with blood or other body fluids from an infected person
- Exposure can occur through sharing of needles for i.v. drug use or through unprotected sex or occupational exposure
- The virus is NOT spread through food or water, casual contact such as holding hands, or coughing or sneezing

Hepatitis C virus (HCV)

- Spread by exposure to contaminated blood, primarily through sharing of needles by i.v. drug users
- Exposure can also occur by sharing personal items contaminated by blood such as razors, through sex with an infected person and via health care occupational exposure

Hepatitis E virus (HEV)

- Endemic in the UK and spread through contaminated food or the faecal-oral route
- Infection is often mild but can cause severe disease in rare cases and during pregnancy
- It is known to cause chronic infection in the immunocompromised

EBV, CMV Syphilis and HIV

- Derby city is a **high prevalence area for HIV**
- **Syphilis cases are increasing nationally and may present with hepatitis**
- **Consider testing for EBV, CMV Syphilis and HIV alongside a hepatitis screen**

Viral Hepatitis Serology Tests

A description of individual tests is below.

Appropriate panels of tests to request according to clinical indication are:

Abnormal LFT follow-up / investigation of chronic hepatitis:

- HBSAg
- HCV IgG

Investigation of acute hepatitis/raised ALT:

- HAV IgM Antibody (only if ALT >500 U/L)
- HBV Surface Antigen
- HCV PCR
- HEV IgM Antibody (only if ALT >100 U/L)
- EBV IgM Antibody
- CMV IgM Antibody
- HIV antigen/antibody
- Syphilis Antibody

Interpretative comments will be added by the laboratory. Further testing will also be added as appropriate according to clinical details such as vaccination status of the patient, whether the test is for past exposure or response to vaccination, whether or not acute hepatitis is suspected and whether the patient is immunocompromised.

Individual tests:

Hepatitis A antibody, IgM (HAV IgM)

- Requesting only appropriate in the investigation of acute viral hepatitis (patient ALT >500 U/L in the previous 6 weeks). Consider in patients with a travel history, MSM or part of an outbreak. Testing is important as vaccination of household contacts is required and immunoglobulin may need issuing to high risk contacts.

Hepatitis B surface Antigen (HBSAg)

- The test detects proteins on the surface of the virus and is the earliest indicator of a current hepatitis B infection. It is usually present even before symptoms appear so is useful for screening those at high risk of infection or who may have been exposed.
- A positive result indicates current infection with hepatitis B virus. If it is the first time the patient is positive then all other confirmatory tests will be added by the lab. A clinical comment and interpretation will also be included.

Hepatitis B surface antibody (HBSAb)

- The presence of hepatitis B surface antibody indicates recovery and immunity from hepatitis B virus infection. Antibodies also develop after successful vaccination.

Hepatitis B core antibody

- If the patient is hepatitis B core antibody positive, they may have current infection or past resolved infection. Therefore, please also request hepatitis B surface antigen

(looking for current infection) and hepatitis B surface antibody (looking for evidence of past cleared infection) to gain a full picture of the patients hepatitis B status. In those with past resolved there is a risk of reactivation if immunosuppressed.

Hepatitis C antibody (HCV IgG)

- The test for hepatitis C antibody cannot distinguish between a current or previous infection, therefore HCV PCR testing is recommended, looking for evidence of current infection/viraemia. This will be added by the lab for all antibody positive samples. If the patient is immunocompromised or you suspect acute infection please request a HCV PCR.

Hepatitis C PCR (HCV PCR)

- If current infection or acute infection is suspected HCV PCR is recommended. A genotype will be added to all first PCR positive samples.

Hepatitis E antibody, IgM (HEV IgM)

- Requesting only appropriate in the investigation of acute viral hepatitis with an ALT >100 U/L (note that if the patient is immunocompromised, ALT can be just at upper limit of normal)
- Hepatitis E PCR will be added if the patient is immunosuppressed (please add these clinical details if known)

EBV IgM and CMV IgM

- This should be considered in all patients with hepatitis and symptoms of fever/sore throat/lymphadenopathy/rash. In this setting an HIV test and syphilis screen should also be requested.

HIV antigen/antibody

- This should be considered in all patients with hepatitis and symptoms or sore throat/lymphadenopathy/rash or any indicator illness as per the British HIV Association and NICE guidance (examples include; pneumonia, hepatosplenomegaly, lymphopenia and lymphadenopathy.
https://www.bhiva.org/file/woZUIBpJXDNMh/GlinesHIVTest08_Tables1-2.pdf)

Contacts

UHDB

Consultant Virologist	10am - 5pm Mon to Fri 01332 788218 (and select option 4) Out of hours - please ask for on-call Microbiologist via Switchboard 01332 340131
Microbiology Manager	01332 788219
Main Microbiology Lab	01332 788218
Serology Lab	01332 788224

Chesterfield

Consultant Virologist	10am - 5pm Mon to Fri 01332 788218 (and select option 4) Out of hours - please ask for on-call Microbiologist via Switchboard 01246 277271
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Secondary Care UHDB	14 th Nov 2025	Dr C Evans	Consultant Virologist (UHDB and CRH)
Secondary Care CRH	14 th Nov 2025	Dr C Evans	Consultant Virologist (UHDB and CRH)
Primary Care	14 th Nov 2025	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	14 th Nov 2025	Ms S Knowles	Consultant Clinical Scientist